Quadratic Equations and Applications

Concept

Quadratic equations can be solved by factoring, square roots, completing the square or a formula.

Formula

$x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$, with $a \neq 0$.

Discriminant

$D > 0$: two real roots; $D = 0$: repeated root; $D < 0$: no real roots.

Zero product

Set the equation equal to zero before setting each factor to zero.

Context

Reject impossible lengths/times; check the vertex lies in the feasible domain.

Example 1

Solve $2x^2 - 5x - 3 = 0$.

Solution

1. Factor: $(2x + 1)(x - 3) = 0$.

2. $2x + 1 = 0$ or $x - 3 = 0 \rightarrow x = -1/2$ or 3.

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Example 2

Solve $x^2 + 4x - 1 = 0$ by completing the square.

Solution

1. $x^2 + 4x = 1 \rightarrow (x + 2)^2 = 5$.
2. Take both roots: $x + 2 = \pm\sqrt{5}$.
3. $x = -2 \pm \sqrt{5}$.

Example 3

Find the maximum of $h(t) = -5t^2 + 20t + 1$ for $t \geq 0$.

Solution

1. Vertex time: $t = -20/[2(-5)] = 2$.
2. $h(2) = -20 + 40 + 1 = 21$. Maximum: 21 at $t = 2$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $x^2 - 9x + 20 = 0$.

2. Solve $2(x - 1)^2 = 18$.

3. Solve $x^2 + 2x + 5 = 0$ over the reals.

4. Find the maximum of $-x^2 + 6x + 4$ for $0 \le x \le 8$.

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Practice answers | check after trying

1. $(x - 4)(x - 5) = 0$, so 4 or 5.

2. $(x - 1)^2 = 9$ gives $x = 1 \pm 3$, so -2 or 4.

3. $D = 4 - 20 = -16$, so no real roots.

4. $-(x - 3)^2 + 13$ has maximum 13 at $x = 3$, which lies in the allowed interval.